

NCA Sponsored Training

Day 3 – Afternoon Session (Lab)

Network Security 2

Ahmed Didouh

Madinah

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rc3.kaust.edu.sa





Outline



1. DNS utilities

- *Host*
- *Dig*

2. Fingerprinting/scanning

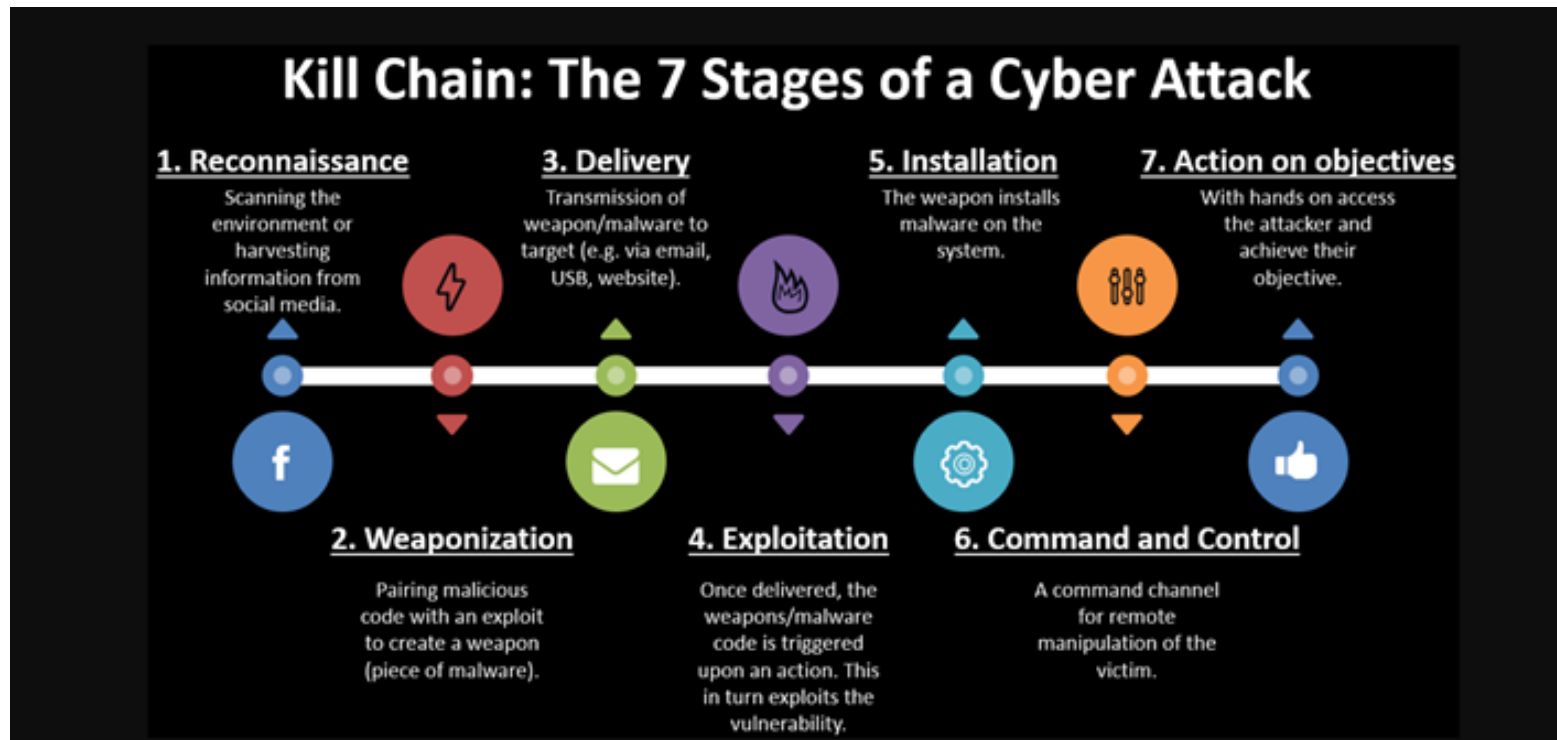
- *nmap*



Lab on Reconnaissance (fingerprinting/scanning)



Reconnaissance (fingerprinting/scanning): First phase of kill chain cyberattack model. It helps the attacker understand the environment around the victim: target addresses, network, architecture, devices, etc.





What is `host` utility?

command used for DNS (Domain Name System) lookup operations, e.g.,:

- find the IP address of a particular domain name or
- find out the domain name of a particular IP address or
- much more DNS configuration details



Host



```
host [-t type] hostname [server]
```

Operation	Command	Example
lookups – mapping names to addresses	host HOST	host google.com
Reverse lookups – mapping addresses to names	host IP_ADDRESS	host 192.168.0.10
lookups at DNS Server	host HOST SERVER	host google.com 8.8.8.8
lookups at DNS Server	host IP SERVER	host 74.125.235.101 8.8.8.8
lookup for an any record	host -t any HOST	host -t any google.com
lookup for an soa record	host -t soa HOST	host -t soa google.com
lookup for an ns record	host -t ns HOST	host -t ns google.com
lookup for an a record	host -t a HOST	host -t a google.com
lookup for an mx record	host -t mx HOST	host -t mx google.com
lookup for an txt record	host -t txt HOST	host -t txt google.com
lookup for an any record -a is equivalent to -v -t ANY	host -a HOST	host -a google.com



What is `dig` utility:

command used to dig deeper in DNS (Domain Name System) lookup operations, e.g.,:

- find the IP address of a particular domain name or
- find out the domain name of a particular IP address or
- much more DNS configuration details



Dig deeper



```
dig [@global-server] [domain] [q-type]
```

Operation	Command	Example
lookups – mapping names to addresses	dig -x HOST	dig google.com
Reverse lookups – mapping addresses to names	dig -x IP_ADDRESS	dig -x 192.168.0.10
lookups at DNS Server	dig @SERVER HOST	dig @8.8.8.8 google.com
lookups at DNS Server	dig @SERVER -x IP	dig @8.8.8.8 -x 74.125.235.101
lookup for an any record (TXT, MX, SOA, A, NS)	dig HOST any	dig google.com any
lookup for an soa record	dig HOST soa	dig google.com soa
lookup for an ns record	dig HOST ns	dig google.com ns
lookup for an a record	dig HOST a	dig google.com a
lookup for an mx record	dig HOST mx	dig google.com mx
lookup for an txt record	dig HOST txt	dig google.com txt
lookup with trace	dig +trace HOST	dig +trace google.com



What is nmap utility?

command (Network Mapper) network detection and security scanning utility,:

- Host discovery (reachable target) with some Ips on the network/Internet
- Scan open/closed ports of a host
- Stealthy scans
- Service version, OS detection
- ..



Nmap - Target specification



```
nmap [scan types] [options] [server]
```

Change IP to target Metasploitable machine as a server

Option	Example	Description
	<code>nmap 192.168.1.3</code>	Scan a specific IP address
	<code>nmap 192.168.1.2 192.168.2.3</code>	Scan specific IP addresses
	<code>nmap 192.168.1.7-254</code>	Scan specific range of IP addresses
	<code>nmap example.com</code>	Scans a domain
-iL	<code>nmap -iL text.txt</code>	Scans a target from a file
-iR	<code>nmap -iR 200</code>	Scans random 200 hosts
-exclude	<code>nmap -exclude 192.168.1.2</code>	Exclude the listed hosts



Set your Firewall



To observe the differences in the following commands, we will use Metasploitable Firewall

1. Go to Metasploitable VM. Our Firewall is UFW, on the terminal check it `man UFW`
2. Run as root (sudo su) and enable Firewall `ufw enable`
3. Go back to KALI and scan some ports (ftp, telnet, ssh) to make sure they are open/close. You can also run `telnet 192.168.1.2` or `ftp ftp 192.168.1.2`
4. If they are open, it means they are not blocked ports, let's block some ports!
5. Go to Metasploitable and block ftp port (21) `ufw deny ftp` (or even block all ports `ufw default deny`)
6. Check `status ufw` and notice the blocked port in Metasploitable
7. Back to KALI to scan and observe the difference
8. Enable port again and recheck: `ufw delete deny ftp` or `ufw allow ftp`



Nmap – Host Discovery



On KALI machine, try these commands against Metasploitable IP and observe the output

Option	Example		Description
-sL	nmap 192.168.1.6-9 -sL		Creates targets List only
-sn	nmap 192.168.1.2/29 -sn		This disables port scans and does host discovery only.
-Pn	nmap 192.168.1.2-5 -Pn		This disables host discovery and allows port scan only.
-PS	nmap 192.168.1.2-5 -PS22-25,80	TCP SYN	TCP SYN ping on port x.
-PA	nmap 192.168.1.2-5 -PA22-25,80	TCP ACK	TCP ACK ping on port x.
-PU	nmap 192.168.1.3-7 -PU53	UDP	Enables UDP ping on port x.
-PR	nmap 192.168.1.2-3/24 -PR	ARP	ARP ping on the local network
-n	nmap 192.168.1.2 -n	DNS	Disables DNS resolution



Nmap – Stealthy Scan



Option	Example	Description	
-sT	nmap 192.168.1.1 -sT	TCP Connect Scan	TCP full handshake connection (not stealthy)
-sS	nmap 192.167.1.2 -sS	TCP SYN Scan	Send SYN. Half-open connection. Bypass Firewall
-sU	nmap 192.168.1.1 -sU	UDP scan	no response means open port
-sA	nmap 192.168.1.1 -sA	TCP ACK Scan	Send ACK. No response means filtered port (Firewall)
-sW	nmap 192.168.1.1 -sX	TCP Xmas scan	Set FIN/URG/PUSH flags. No response means open/filtered
-sM	nmap 192.168.1.1 -sM	TCP Maimon scan	Set FIN/ACK flags. No response means open/filtered port



Nmap – Port Specification



Option	Example	Description
-p	nmap 192.168.1.9 -p 27	Scan a specific port
-p	nmap 192.168.1.9 -p 27-100	Scan a port range
-p	nmap 192.168.1.9 -p U:53,T:27-40,80	Scans multiple TCP and UDP ports
-p-	nmap 192.168.1.9 -p-	Scan all ports
-p	nmap 192.168.1.9 -p http,https	Scans based on the service name
-F	nmap 192.168.1.9 -F	Scan 100 ports in fast manner
-top-ports	nmap 192.168.1.9 -top-ports 1015	Scans the top “x” ports



Nmap – Port Specification



Other utilities

- whois
- Traceroute
- Nslookup



Backlog



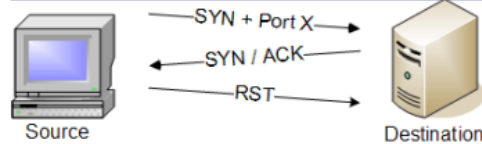


Nmap visuals

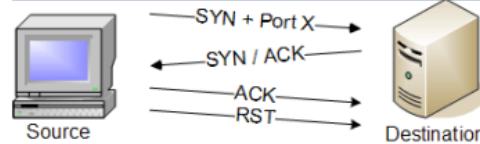


Identifying Open Ports with Nmap

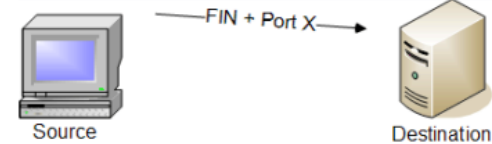
TCP SYN SCAN (-sS)



TCP connect() SCAN (-sT)



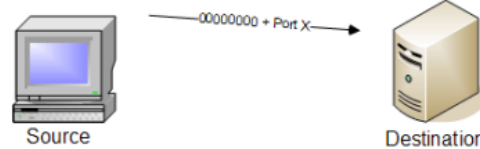
TCP FIN SCAN (-sF)



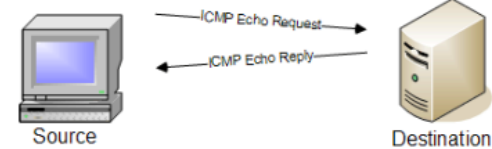
TCP XMAS TREE SCAN (-sX)



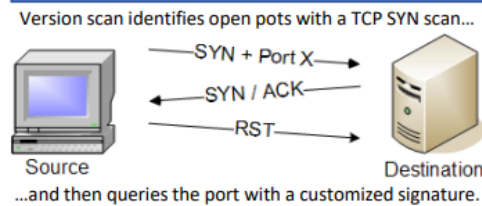
TCP NULL SCAN (-sN)



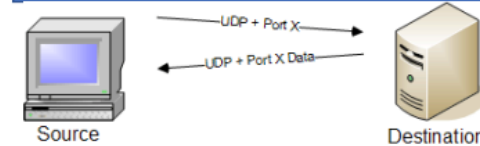
TCP PING SCAN (-sP)



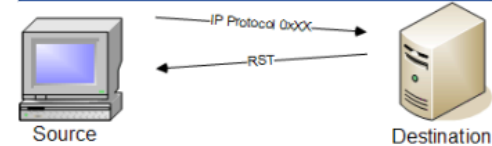
VERSION DETECTION SCAN (-sV)



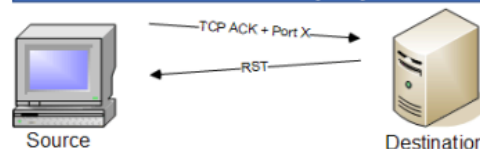
UDP SCAN (-sU)



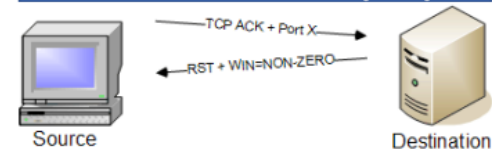
IP PROTOCOL SCAN (-sO)



TCP ACK SCAN (-sA)



TCP WINDOW SCAN (-sW)



Do you have any questions?



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